

Capstone Project

Customer Churn Analysis & Retention Strategy

1. Project Overview and Objectives

Project Overview

This project performs end-to-end data analysis on customer churn using statistical methods and machine learning. The goal is to identify key factors influencing customer attrition and build predictive models to detect high-risk customers. Insights from exploratory analysis and modeling are used to generate actionable strategies to reduce churn and protect business revenue.

Objectives

- To analyze customer behavior and subscription data to identify churn patterns.
- To identify key factors such as tenure, contract type, and monthly charges affecting retention.
- To build machine learning models to predict customers likely to churn.
- To generate actionable insights and strategies to reduce churn and protect revenue.

2. Setup and Installation Instructions

Prerequisites

- Python 3.10+
- Jupyter Notebook

Installation Steps

1. Download or clone the project repository from GitHub.
2. Install required libraries:

```
pip install -r requirements.txt
```

3. Place dataset inside /data folder.
5. Launch Jupyter Notebook and open capstone_analysis.ipynb.

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3. Code Structure Explanation

CAPSTONE_PROJECT-REAL_WORLD_BUSINESS_ANALYSIS

```
|
|— data
|   |— raw_data.csv
|   |— cleaned_data.csv
|   |— customer_risk_scores.csv
|
|— images
|   |— (visualization screenshots)
|
|— notebooks
|   |— 1_data_cleaning.ipynb
|   |— 2_eda.ipynb
|   |— 3_analysis.ipynb
|
|— presentation
|   |— business_presentation.pptx
|
|— reports
|   |— executive_summary.docx
|   |— executive_summary.pdf
|   |— technical_report.docx
|   |— technical_report.pdf
|
|— capstone_analysis.ipynb
|— README.md
|— requirements.txt
|— .gitignore
```

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File Descriptions

- data/raw_data.csv
Original customer churn dataset containing customer demographics, subscription details, and churn status.
- data/cleaned_data.csv
Dataset after cleaning and feature engineering.
- data/customer_risk_scores.csv
Generated output containing predicted churn probabilities used for identifying high-risk customers.
- Images/
Folder containing visualization screenshots such as churn distribution, correlation matrix, ROC curve, and feature importance charts.
- notebooks/1_data_cleaning.ipynb
Performs data loading, cleaning, and feature creation.
- notebook/2_eda.ipynb
Exploratory data analysis and visualization of churn patterns.
- notebook/3_analysis.ipynb
Machine learning models for churn prediction and evaluation.
- capstone_analysis.ipynb
Complete end-to-end analysis including modeling, tuning, and business insights.
- presentation/business_presentation.pptx
Business-focused presentation summarizing insights, visualizations, and recommended strategies to reduce customer churn.
- reports/executive_summary.pdf
One-page summary highlighting the business problem, key findings, and strategic recommendations.

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- reports/technical_report.pdf
Detailed report explaining the methodology and analysis.
- requirements.txt
Specifies all Python dependencies needed to run the project.
- README.md
Project documentation including overview, setup instructions, and usage guide.

4. Screenshots of Working Application

The included screenshots demonstrate:

- The contract analysis shows that month-to-month customers have the highest churn rate, indicating short-term contracts increase churn risk.
- The monthly charges distribution shows that customers with higher charges are more likely to churn.
- The correlation matrix indicates a strong negative relationship between Tenure and Churn (-0.51), meaning newer customers churn more frequently.
- The model evaluation shows high predictive performance (Accuracy: 95%, ROC-AUC: 0.99), confirming effective churn prediction.

Logistic Regression Report				
	precision	recall	f1-score	support
0	0.98	0.95	0.96	84
1	0.78	0.88	0.82	16
accuracy			0.94	100
macro avg	0.88	0.91	0.89	100
weighted avg	0.94	0.94	0.94	100

Fig 4.1. Logistic Regression Classification Report

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Random Forest Report					
	precision	recall	f1-score	support	
0	0.94	1.00	0.97	84	
1	1.00	0.69	0.81	16	
accuracy			0.95	100	
macro avg	0.97	0.84	0.89	100	
weighted avg	0.95	0.95	0.95	100	

Fig.4.2. Random Forest Classification Report

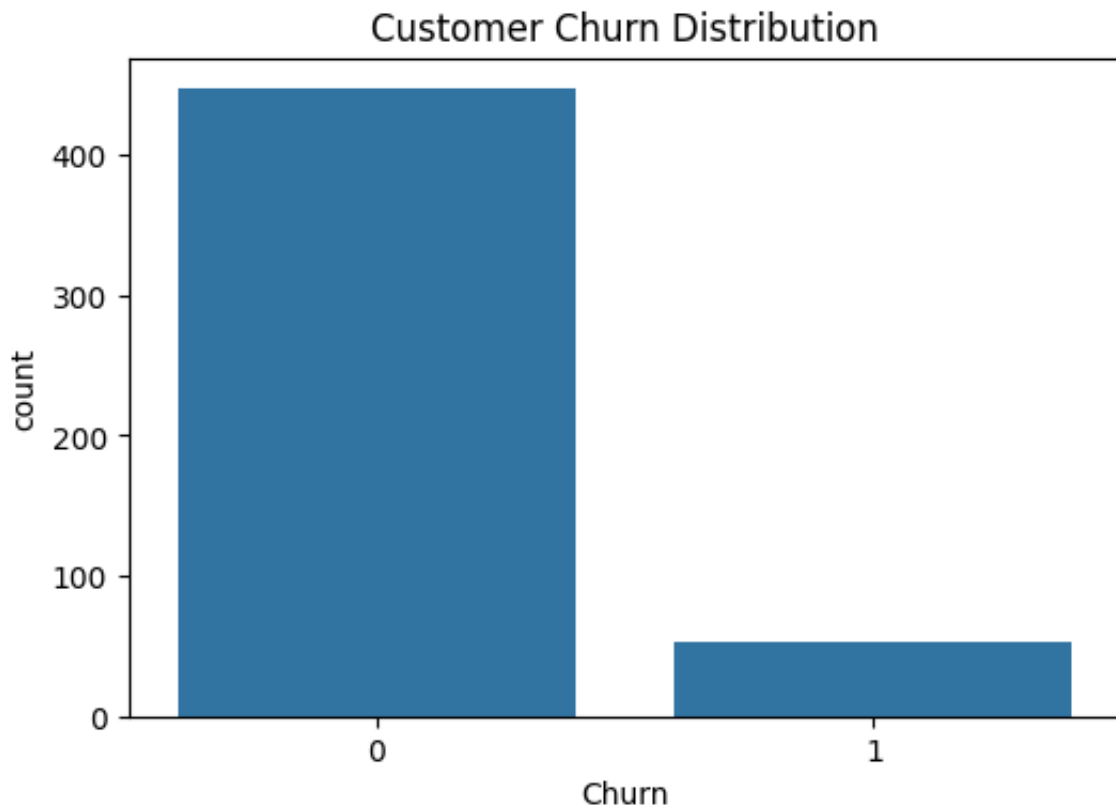


Fig.4.3. Churn Distribution

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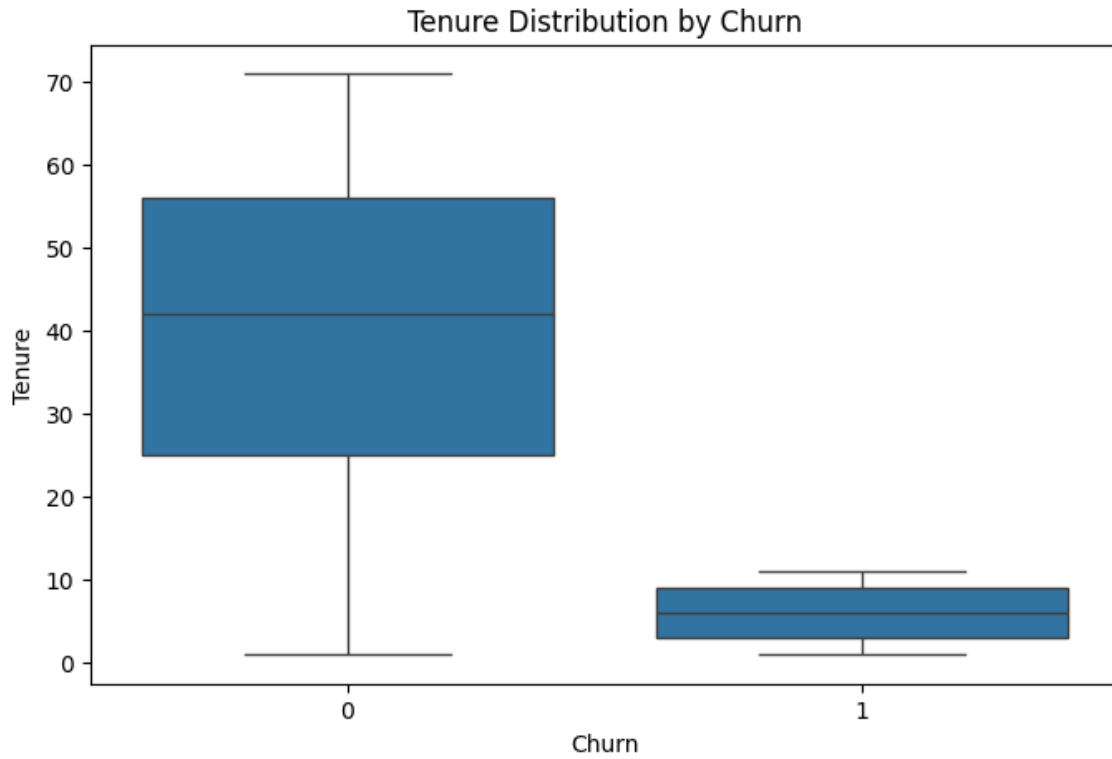


Fig.4.4. Distribution of Tenure by Churn

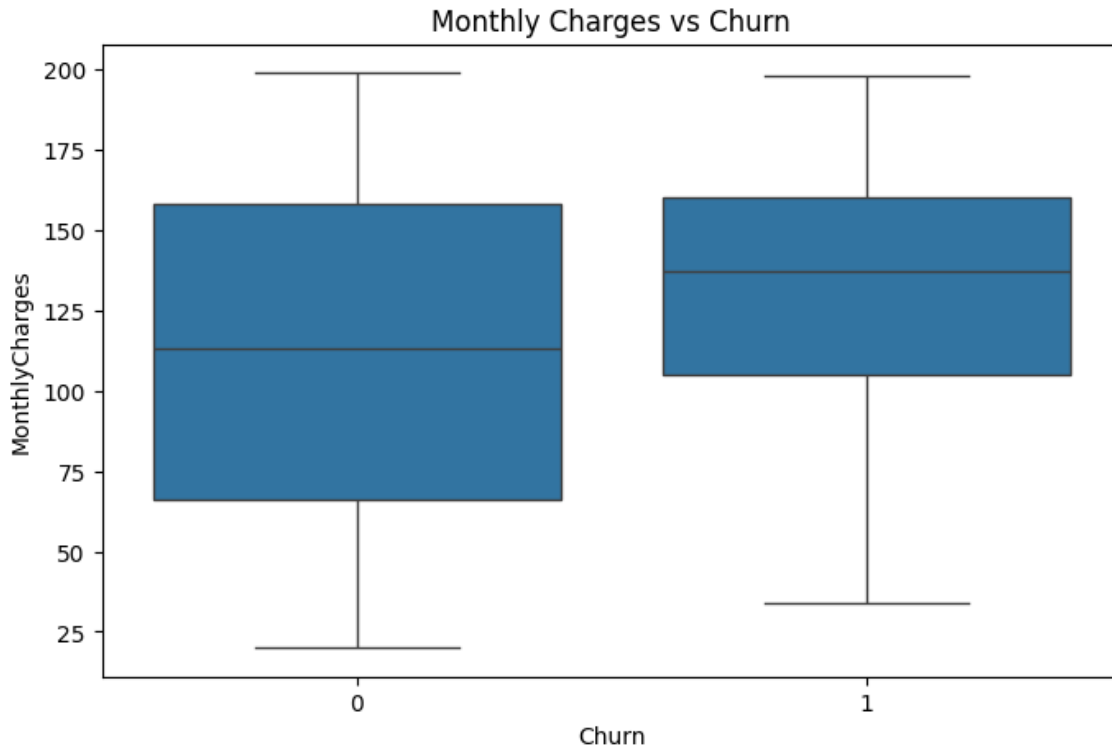


Fig.4.5. Distribution of Monthly Charges by Churn Status

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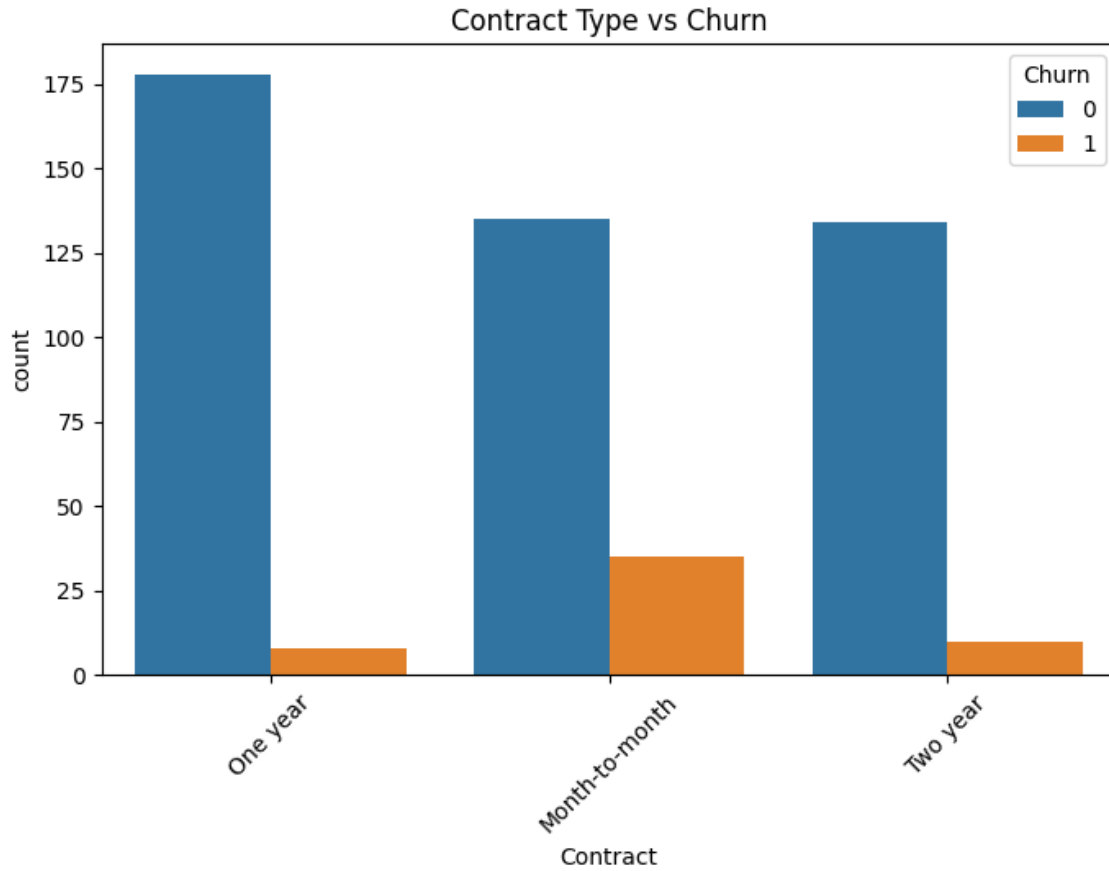


Fig.4.6. Contract Type vs Customer Churn

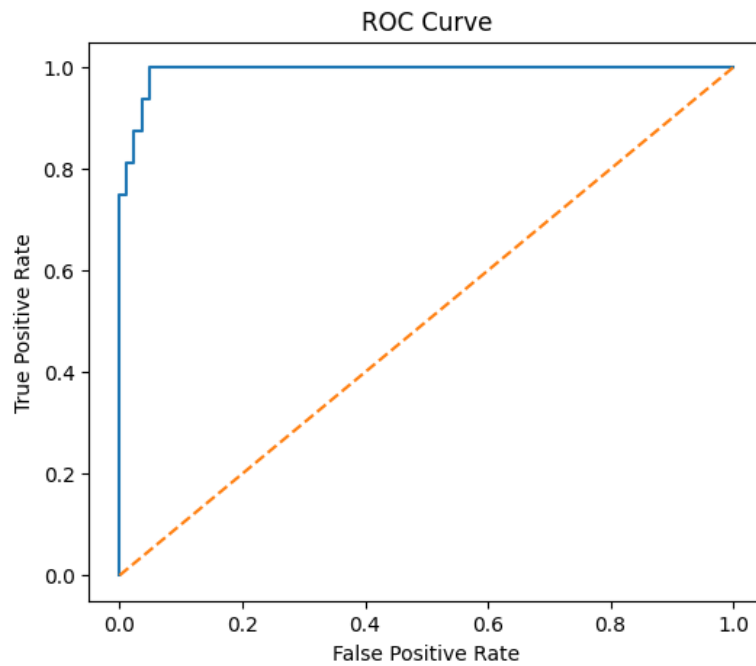


Fig.4.7. ROC Curve for Churn Prediction Model

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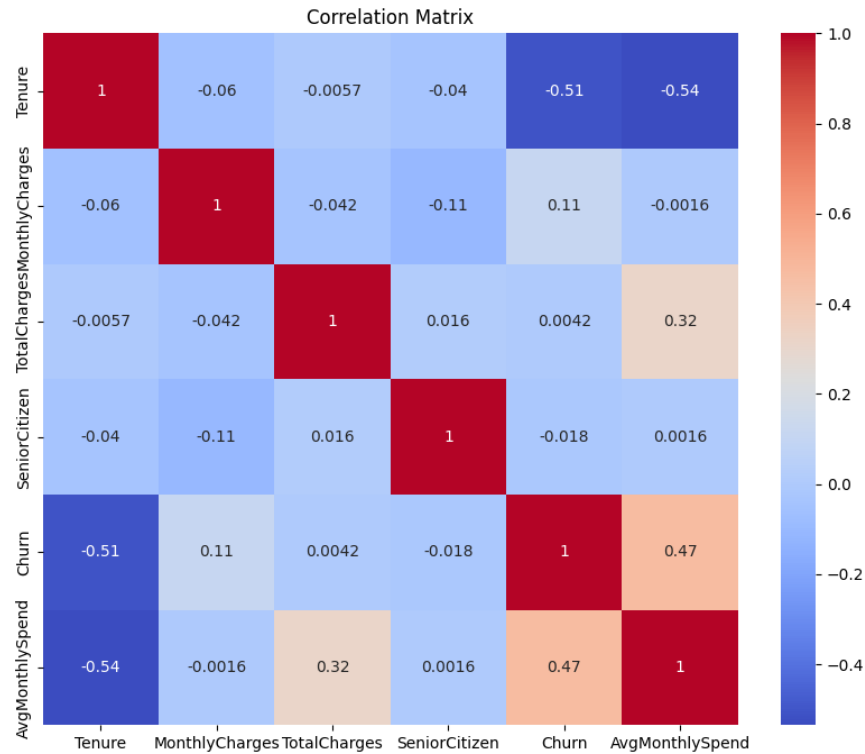


Fig.4.8. Correlation Matrix of Key Customer Features

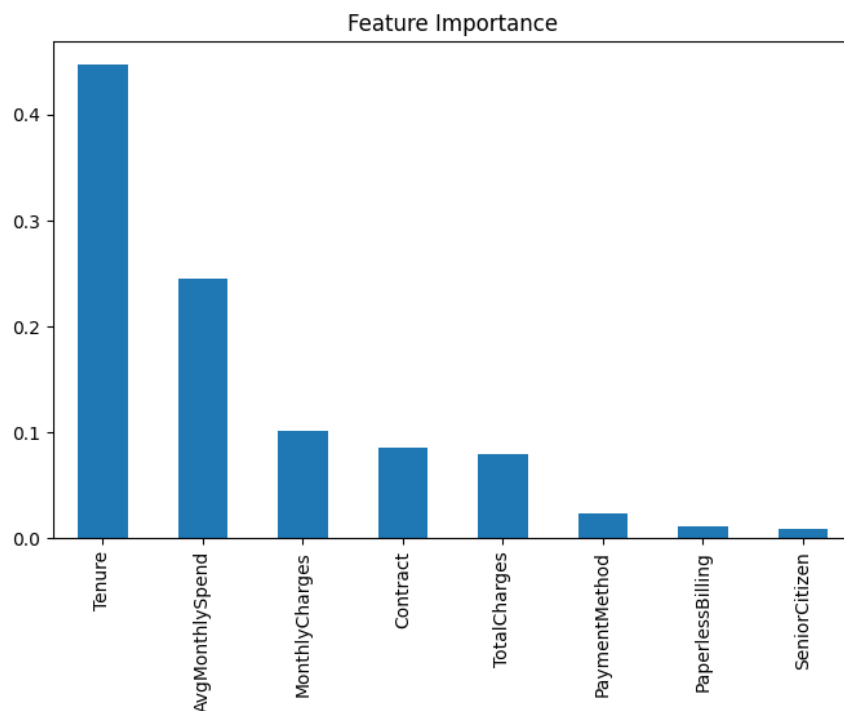


Fig.4.9. Feature Importance from Random Forest Model

These results validate that the analysis pipeline and predictive models were successfully implemented and provide insights for improving customer retention.

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5. Technical Requirements Fulfillment Explanation

Category	Technical Requirement	How It Was Implemented
Algorithm	Data loading and preprocessing	Customer churn dataset loaded using Pandas, validated for minimum size, and datatypes standardized.
Algorithm	Data cleaning logic	TotalCharges converted to numeric, missing values filled with median, and duplicate records removed.
Algorithm	Feature engineering	New feature AvgMonthlySpend created using TotalCharges and Tenure to capture customer spending behavior.
Algorithm	Exploratory data analysis	Churn distribution, tenure trends, and contract relationships analyzed using Seaborn boxplots and countplots.
Algorithm	Statistical validation	T-Test used for tenure differences between churn groups and Chi-Square test applied for contract vs churn relationship.
Algorithm	Predictive modeling	Logistic Regression and Random Forest classifiers implemented for churn prediction using Scikit-learn.
Algorithm	Model optimization	Cross-validation and GridSearchCV used to tune Random Forest hyperparameters and improve model performance.

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Category	Technical Requirement	How It Was Implemented
Data Structure	Tabular data handling	Pandas DataFrames used to store, transform, and analyze structured customer data.
Data Structure	Feature storage	Encoded categorical features and scaled numerical features stored in arrays for model training.
Architecture	Modular notebook design	Workflow divided into separate notebooks: data cleaning, EDA, analysis, and capstone modeling.
Architecture	Reproducible pipeline	Data preprocessing, feature engineering, model training, and evaluation executed in a sequential analytical pipeline.
Output	Visualization outputs	Professional plots including correlation heatmap, churn distribution, ROC curve, and feature importance generated.
Output	Model evaluation metrics	Performance measured using precision, recall, F1-score, accuracy, and ROC-AUC metrics.
Output	Business insights	Key churn drivers such as short tenure and month-to-month contracts identified with supporting analysis.

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Category	Technical Requirement	How It Was Implemented
Output	Revenue risk analysis	Estimated annual revenue at risk calculated for high-probability churn customers and exported as risk scoring file.

6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained practical experience in performing end-to-end data analysis using Python.
- Learned data cleaning, preprocessing, and feature engineering techniques.
- Developed skills in exploratory data analysis and data visualization using Matplotlib and Seaborn.
- Applied machine learning and statistical methods to identify churn patterns and key business drivers.

7. Conclusion

This project demonstrated how data analysis and machine learning can identify customers at risk of churn. By applying EDA, statistical testing, and predictive modeling, key factors influencing churn were discovered. The insights enable businesses to implement targeted retention strategies and reduce potential revenue loss.